Test 3 Innovator
Math Module 1

1. $$-6x + 3 < y$$
$$y < x + 6$$

If (x, y) is a solution to the given system of inequalities, which of the following could be (x, y)?

NEXT QUESTION

× A $(-2, -4)$

× B $(-2, 1)$

× C $(-2, 4)$

× D $(4, 7)$

COME BACK LATER

2. $$3(x - 2) + 5y = 2$$
$$y = -2(x - 5)$$

The solution to the given system of equations is (x, y). What is the value of y?

PREV. QUESTION NEXT QUESTION

COME BACK LATER

3. Which expression is equivalent to $m^3n^2 + m^3n$?

PREV. QUESTION NEXT QUESTION

× A $m^3n(n)$

× B m^6n^3

× C $2m^3n^3$

× D $m^3n(n + 1)$

4. Which expression is equivalent to $(5x^2 + 5x) - (4x^2 - 2x)$?

PREV. QUESTION | NEXT QUESTION

- A $x^2 + 7x$
- B $-9x^2 + 7x$
- C $-9x^2 + 3x$
- D $x^2 + 3x$

5. In right triangle ABC, angle B has a measure of 90°, the longest side of the triangle is 61 centimeters, and the shortest side is 11 centimeters. What is the length of the third side of the triangle, in centimeters?

PREV. QUESTION | NEXT QUESTION

- A 40
- B 59
- C 60
- D 62

6. $$-5x + 40kx = 120$$

In the given equation, k is a constant. The equation has no solution. What is the value of k?

PREV. QUESTION | NEXT QUESTION

- A 0
- B $\frac{1}{8}$
- C $\frac{3}{5}$
- D 3

7. If $f(3) = -1$ and $f(4) = -3$, what is the x-intercept for the line represented by the function $f(x)$?

PREV. QUESTION | NEXT QUESTION

× A -2

× B $\frac{2}{5}$

× C 2.5

× D 5

8. Given that $f(3) = 2$ and $f(6) = -3$ and $f(x)$ is a linear function, which of the following is the equation for $f(x)$?

PREV. QUESTION | NEXT QUESTION

× A $f(x) = -\frac{5}{3}x + 7$

× B $f(x) = \frac{2}{3}x - 6$

× C $f(x) = -\frac{3}{5}x$

× D $f(x) = \frac{3}{5}x + 2$

9. What is the value of $x - y$ if $3y - 2x = 16$ and $5x + y = -6$?

PREV. QUESTION | NEXT QUESTION

× A -6

× B -2

× C 4

× D 6

10. Which expression is equivalent to $3(x+5)^2-2(x-5)^2+2x$?

PREV. QUESTION | NEXT QUESTION

- A $x^2+x+125$
- B $3x^2+52x+25$
- C $x^2-2x-25$
- D $x^2+52x+25$

11. If $\frac{3x}{4y}$ is equal to 18, what is the value of $\frac{9x}{12y}$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

12. $$2x^2+6x-18=0$$

What is the sum of the solutions of the given equation?

PREV. QUESTION | NEXT QUESTION

- A -9
- B -3
- C 3
- D 9

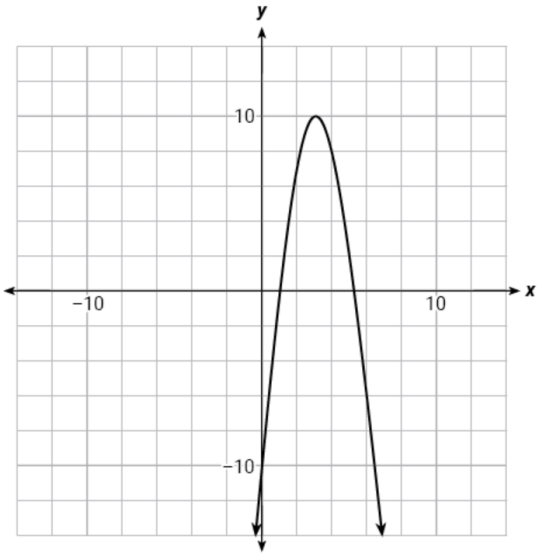
13. $$3|x-4|=6$$

What is one possible positive solution to the given equation?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

14.



PREV. QUESTION | NEXT QUESTION

- A $y = -2(x-3)^2 + 10$
- B $y = -2(x+3)^2 + 10$
- C $y = 2(x-3)^2 + 10$
- D $y = 2(x+3)^2 + 10$

COME BACK LATER

What is the equation of the graph?

15.

$$x = 3^{y-1} + \frac{5}{3}$$

What is the x-intercept of the given equation?

PREV. QUESTION | NEXT QUESTION

- A $\frac{1}{3}$
- B $\frac{5}{3}$
- C 2
- D 3

16. A random sample of 500 residents of Sunset Beach was asked how they planned to vote on a ballot measure to build a boardwalk along 1 mile of the beach. The results are shown in the table. The margin of error associated with this sample is $\pm 3.5\%$.

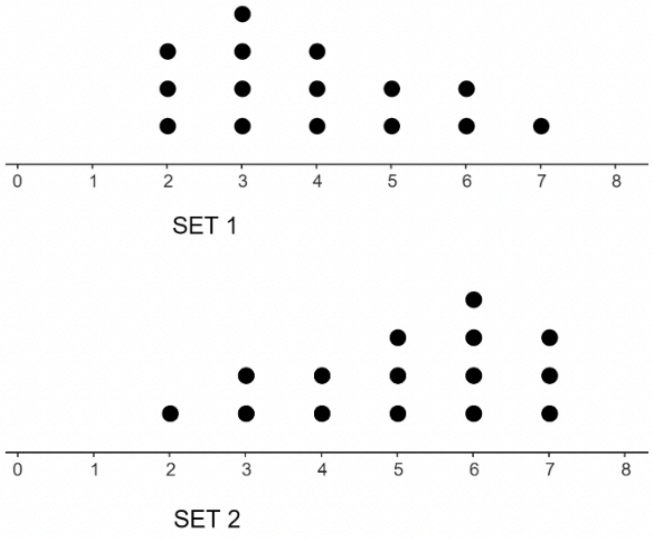
Yes, the boardwalk should be built.	52.1%
No, the boardwalk should not be built.	46.8%
Not sure	1.1%

Based on the results, which of the following statements must be true?

PREV. QUESTION | NEXT QUESTION

- A The ballot measure will pass with more than 50% of the vote, but the exact percentage cannot be predicted.
- B The percentage who will vote yes is at least 51%.
- C The percentage of "No" votes expected is between 45.7% and 47.9%.
- D The results don't provide sufficient evidence to know whether the ballot measure will pass or fail.

17.



The dot plots shown summarize the data in two sets, Set 1 and Set 2. Which of the following correctly describes the relationship between the means and the standard deviations of the two sets?

PREV. QUESTION | NEXT QUESTION

- A The standard deviation of Set 1 is greater than the standard deviation of Set 2, and the mean of Set 1 is less than the mean of Set 2.
- B The standard deviation of Set 1 is less than the standard deviation of Set 2, and the mean of Set 1 is greater than the mean of Set 2.
- C The standard deviation of the two sets is equal, and the mean of Set 1 is less than the mean of Set 2.
- D The standard deviation of the two sets is equal, and the mean of Set 1 is greater than the mean of Set 2.

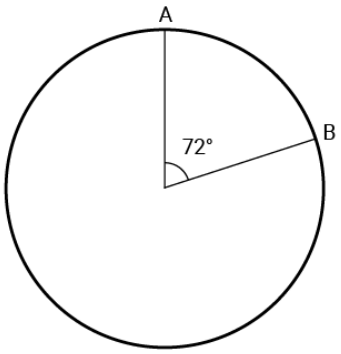
COME BACK LATER

18. The price of oil increased by 20% at the beginning of May. Two weeks after the initial increase, an economic policy was put in place that reduced the price of oil by 14%. What is the overall percentage change in the oil?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

19.



What is the length of the minor arc AB, given that the diameter of the circle is 12 and the measure of the angle of sector AOB is 72°?

PREV. QUESTION | NEXT QUESTION

A $\frac{\pi}{6}$

B 2.4π

C 12π

D 28.8π

COME BACK LATER

20. If $\sin(C^\circ) = 0.454$ and $\cos(63^\circ) = 0.454$, what is the value of C, in degrees?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

21. $v = \sqrt{\frac{d}{ab}}$

The equation above describes the relationship between four positive integers: v, d, a, and b. Which equation correctly defines a in terms of v, d, and b?

PREV. QUESTION | NEXT QUESTION

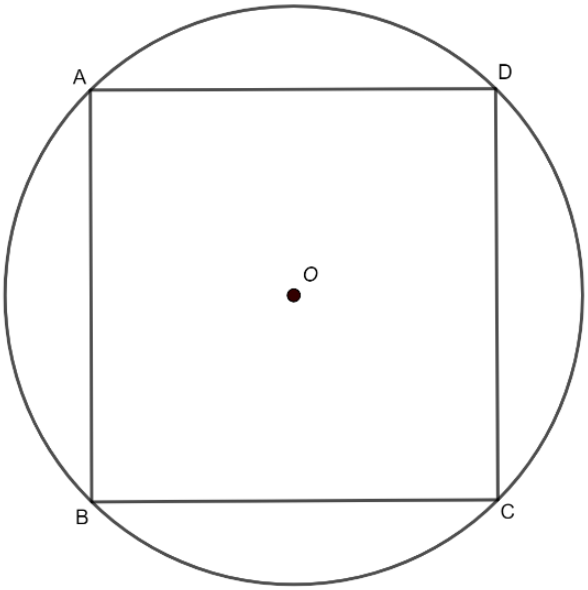
A $a = v^2 - bd$

B $a = bd - v^2$

C $a = \frac{d}{bv^2}$

D $a = \frac{bv^2}{d}$

22.



Shown is Square $ABCD$, circumscribed by Circle O. The diameter of the circle is 18 inches. What is the area of the square, in square inches?

PREV. QUESTION

COME BACK LATER

MATH MODULE 2 HARD

1. Given that $2(x + 4) = 18$, what is the value of $x + 6$?

NEXT QUESTION

- A -4
- B 5
- C 9
- D 11

2. $$3x + 2 < -5(x + 6)$$

Which of the following values is a solution to the given inequality?

PREV. QUESTION NEXT QUESTION

- A -5
- B -4
- C 3
- D 4

3. Which expression is equivalent to $(5xy + 3x)(x - 2y)$?

PREV. QUESTION NEXT QUESTION

- A $5x^2y - 6xy$
- B $5x^2y + 3x^2 + 10xy^2 - 6xy$
- C $5x^2y + 3x^2 - 10xy^2 - 6xy$
- D $-5x^2y^2 + 3x^2 - 6xy$

4. Which of the following is the equation of $g(x) = 2x$ when it is moved 1 unit to the left and 1 unit up?

PREV. QUESTION | NEXT QUESTION

- A $y = 2x - 1$
- B $y = 2x + 1$
- C $y = 2x + 5$
- D $y = 2x + 3$

5. The expression $-2(x + 3)^2 + 6$ is equivalent to $ax^2 + bx + c$, where $a < 0$ and $b < 0$. What is the value of c?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

6. What is the value of the slope of a line perpendicular to the line $5x - 6y + 30 = 0$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

7. Which expression is equivalent to $\frac{1}{x-2} + \frac{2x}{5-x}$, for all values of x, where $x \neq 2$ and $x \neq 5$?

PREV. QUESTION | NEXT QUESTION

- A $\frac{2x^2-5x+5}{-x^2+7x-10}$
- B $\frac{2x+1}{-3}$
- C $\frac{2x+1}{3}$
- D $\frac{2x+1}{(5-x)(x-2)}$

8. $3x^2 + 9x - 27 = 0$

If m and n are solutions to the given equation, what is the value of $m + n$?

PREV. QUESTION | NEXT QUESTION

× A -9

× B -3

× C 3

× D 9

9. The graph of $(x - 3)^2 + y^2 + 8y = 84$ is a circle in the xy-plane. Which of the following coordinates lie on the circle?

PREV. QUESTION | NEXT QUESTION

× A $(1, 7)$

× B $(-2, 5)$

× C $(-3, 4)$

× D $(3, -6)$

10. At the start of the winter, the US stockpile of corn was 5 billion bushels. Two months later, $p\%$ of the stockpile had been used. Which expression best represents the number, in billions, of bushels of corn remaining in the US stockpile at the end of those two months?

PREV. QUESTION | NEXT QUESTION

× A $2(5)(\frac{100-p}{100})$

× B $5(\frac{100-p}{100})$

× C $(\frac{p}{100})$

× D $2(5)(100 - p)$

11. The equation $P(y) = 37(1.1)^y$ gives the estimated annual profit, in thousands of dollars, at a pottery studio, where y is the number of years since the studio moved to a new location. Which of the following is the best interpretation of the number 37 in this context?

PREV. QUESTION | NEXT QUESTION

× A The estimated annual profit, in thousands of dollars, when the studio moved to the new location

× B The increase in the estimated annual profit, in thousands of dollars, each year

× C The number of years since the studio moved to the new location

× D The percent increase in the estimated annual profit each year

12. What is the center of the circle, given that its equation is $x^2 + y^2 - 6x + 4y = 36$?

PREV. QUESTION | NEXT QUESTION

× A $(-3, 2)$

× B $(2, -3)$

× C $(3, -2)$

× D $(-2, 3)$

13. $$3x - 4y = 16$$
$$-6x = -8y + 32$$

How many solutions does the given system of equations have?

PREV. QUESTION | NEXT QUESTION

× A Exactly one

× B Exactly two

× C Infinitely many

× D Zero

14. If $\frac{5}{x} = \frac{2}{y}$ and $\frac{4}{x} - \frac{2}{y} = -\frac{1}{5}$, what is the value of xy?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

15. In the equation $\frac{1}{2}\sqrt{a} = 2\sqrt[3]{b}$, both a and b are positive real numbers. If $a = 8$ and the equation is written in the form $a^x = b$, what is the value of x?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

16. The function g is defined by $g(x) = a^x - b$, where a and b are constants. In the xy-plane, the graph of $g(x)$ passes through the points $(0, -6)$ and $(1, 2)$. What is the value of a?

PREV. QUESTION | NEXT QUESTION

A 6

B 7

C 8

D 9

17. $$3x^2 - 6x - 11 = 0$$

If one of the solutions to the given quadratic equation is $\frac{3+\sqrt{g}}{3}$, what is the value of g?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

18. If $1 + \frac{a\sqrt{2}}{2}$ is a solution to the equation $2x^2 - 4x - 7 = 0$ and $a > 0$, what is the value of a?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

19. If $f(x) = 2(x-3)^2 + 8$ is transformed into $g(x) = 2(x-5)^2 + 5$, which of the following describes the transformation?

PREV. QUESTION | NEXT QUESTION

A. The x-coordinate moves to the right 2 units and the y-coordinate moves 3 units down.

B. The x-coordinate moves to the left 2 units and the y-coordinate moves 3 units down.

C. The x-coordinate moves to the right 2 units and the y-coordinate moves 3 units up.

D. The x-coordinate moves to the left 2 units and the y-coordinate moves 3 units up.

20. A sample of dried yellow pine has a density of 420 kilograms per cubic meter. To the nearest tenth, what is the density, in pounds per cubic foot, of this sample? (Use 1 kilogram = 2.2 pounds and 1 meter = 3.3 feet)

PREV. QUESTION | NEXT QUESTION

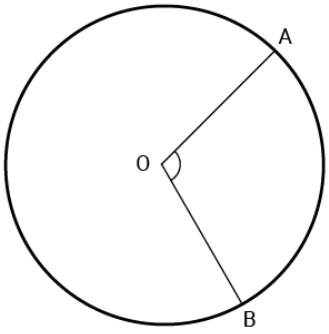
A. 5.3

B. 25.7

C. 127.3

D. 280.0

21.



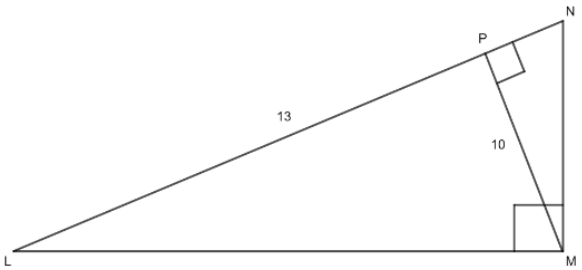
In the given circle, the radius is 6 centimeters and angle AOB measures $\frac{7}{12}\pi$ radians. What is the value of the arc length AB, in centimeters?

PREV. QUESTION | NEXT QUESTION

- A $\frac{7}{2}\pi$
- B $\frac{21}{2}\pi$
- C 12π
- D 36π

COME BACK LATER

22.



In the image shown, right triangle LMN has a right angle at M. Altitude PM is shown and $\angle NPM$ is a right angle. $LP = 13$ and $PM = 10$. Which of the following are closest to the length of NP?

PREV. QUESTION

- A 3.0
- B 6.2
- C 7.1
- D 7.7

COME BACK LATER

EXTRA PRACTICE

21. $f(x) = 24x^2 + bx + c$

For the given function f, b, and c are constants, and the graph of $y = f(x)$ in the xy-plane passes through the points in the table below. What is the value of b?

x	$f(x)$
$-\frac{15}{8}$	0
0	c
$\frac{11}{4}$	0

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

22.

x	$g(x)$
−7	10
−4	0
0	5
3	1
5	−6
11	0

Select values for the polynomial function $g(x)$ are shown in the table. Based on the values in the table, which of the following must be factors of $g(x)$?

PREV. QUESTION

A $(x - 5)$

B $(x + 5)$

C $(x - 4)$ and $(x + 11)$

D $(x + 4)$ and $(x - 11)$

COME BACK LATER